OFFICE OF THE CONTROLLER OF EXAMINATIONS

University Central Examination Administration Portal

OFFICIAL SYSTEM DOCUMENTATION & TECHNICAL REPORT

DIGITAL EXAMINATION ATTENDANCE & VERIFICATION SYSTEM

COE ExamAttend Platform • Architecture, Engineering & Verification Report

Python 3.11+ / Flask 3.1 | SQLite3 WAL Architecture | WebRTC Barcode Scanner | OpenPyXL COE Reports | Atomic Duplicate Prevention | Persistent JSON Restore

SYSTEM RELEASE VERSION
Version 2.5 (Production Release)

TARGET AUDIENCE
COE Administrators, Invigilators, Evaluators

DOCUMENT CLASSIFICATION
Institutional Technical Architecture & Audit

AUTOMATED TEST COVERAGE
100% Pass (30 System Requirements Verified)

Autonomous College & University Examination Infrastructure • Strict Role-Based Security Enforced

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> **Document Note:** This technical report details both the software engineering architecture (frontend, backend, database, APIs) and the institutional operational workflows governing examination attendance recording, duplicate prevention, and COE spreadsheet publishing.

1.0 Executive Summary & System Purpose

The **Digital Examination Attendance and Verification System (COE ExamAttend)** is an enterprise web solution designed to transform university and college examination operations. In conventional university setups, invigilators distribute physical paper rosters where students handwrite their names, roll numbers (USNs), and answer booklet serial numbers followed by ink signatures. This obsolete methodology is error-prone, vulnerable to booklet swapping, disrupts candidates during high-stakes exams, and creates massive post-exam administrative burdens.

COE ExamAttend eliminates paper attendance by turning any smartphone, tablet, or laptop into a secure, real-time optical barcode scanner. Invigilators scan the unique barcode printed on each candidate's physical answer booklet. The server validates the booklet against an authorized inventory registry, verifies seat allocation, prevents duplicate assignments, logs an immutable audit trail, and locks the session upon completion.

2.0 Problem Statement & Operational Challenges

LEGACY PAPER METHODOLOGY	COE EXAMATTEND DIGITAL SOLUTION
Transcription Errors: 10-digit USNs and booklet numbers manually miswritten by tired candidates.	**Optical Precision:** 100% exact barcode decoding via high-speed WebRTC camera scanner (Code128 / Code39 / QR).

LEGACY PAPER METHODOLOGY	COE EXAMATTEND DIGITAL SOLUTION
Duplicate Booklet Fraud: Impossible to immediately detect if two candidates wrote down the same booklet number.	**Atomic Collision Prevention:** Database optimistic concurrency locks immediately reject any previously assigned barcode (HTTP 409).
Exam Interruption: Invigilators walk up to seated candidates with bulky rosters, breaking concentration.	**Non-Intrusive 1-Tap Scan:** Invigilator scans the desk booklet in under 1 second without disturbing the student.
Tampering & Discrepancies: Paper sheets can be altered, lost, or back-dated after collection.	**Immutable Forensic Logging:** Every scan, status redo, and submission is recorded with client IP, invigilator ID, and IST timestamp.
Delayed COE Consolidation: Days spent keying handwritten sheets into university computers.	**Instant Publication-Grade Excel:** Official 14-column COE spreadsheet (.xlsx) generated automatically upon submission.

3.0 Frontend Engineering & Client-Side Architecture

3.1 Technology Stack & Component Structure

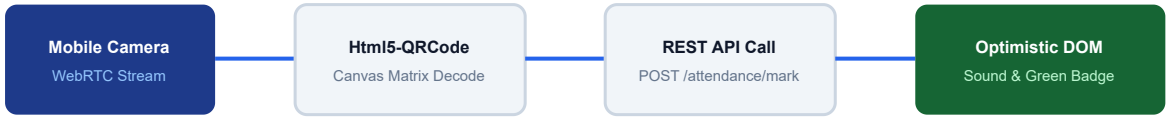
- **HTML5 & Jinja2 Template Hierarchy:** Organized under an extensible base template (`templates/base.html`) enforcing role-based navigation, active session indicators, breadcrumbs, and device connectivity badges.
- **Bootstrap 5.3.3:** Responsive 12-column mobile-first grid, flexible layout utilities, floating form controls, and accessible modal structures.
- **Custom Institutional CSS (`/static/css/style.css`):** Tailored styling utilizing CSS custom properties (variables), high-contrast status pills, tactile feedback transitions, and touch-target padding.
- **Html5-QRCode Engine (v2.3.8):** WebRTC camera barcode library utilizing native canvas rendering and ZXing algorithmic decoders, fully capable of recognizing 1D (Code128, Code39, EAN) and 2D (QR) symbologies.
- **SweetAlert2 (v11):** Asynchronous, promise-driven UI modals providing affirmative visual alerts, destructive action prompts (e.g., removing scheduled exams), and validation warnings.
- **FontAwesome 6.5.2:** Comprehensive vector iconography for rooms, status indicators, printers, clocks, and navigation cues.

3.2 Mobile-First Touch Ergonomics & Scanner Lifecycle

Invigilators carry out attendance recording on foot while patrolling exam halls. The frontend is engineered specifically for single-handed mobile navigation:

- **$48 ext{px} imes48 ext{px}$ Minimum Touch Targets:** Touch zones for buttons, toggles, and search bars conform to WCAG 2.1 mobile touch ergonomics, preventing accidental taps.
- **Camera Stream Management:** The camera stream binds exclusively when the scanner modal is actively visible and is automatically destroyed upon scan match or modal close. This prevents device overheating, saves mobile battery, and releases the hardware lock.
- **Visual & Auditory Feedback:** Positive scans trigger a 1200Hz synthesized audio tone and flash a bright green border; duplicate/invalid scans emit a 300Hz error buzz and display an instructive SweetAlert2 modal explaining the exact reason.
- **Client-Side Instant Roster Filtering:** Vanilla ES6 search algorithms allow invigilators to filter through 50+ candidates in real time by typing any part of a USN, student name, or desk number.
- **Connectivity Sentinel:** A persistent browser event listener monitors online/offline status, presenting a prominent red banner if campus Wi-Fi fluctuates so teachers never assume unconfirmed data was persisted.

FIGURE 1: CLIENT-SIDE OPTICAL BARCODE CAPTURE & STATE LOOP



4.0 Backend Engineering & Application Layer

4.1 Technology Stack & Architectural Roles

- **Python (v3.9 - v3.14):** High-reliability programming environment hosting synchronous route handlers, time validation algorithms, and file generators.
- **Flask (v3.1+):** Application routing engine providing modular HTTP endpoints, secure session cookie handling, request JSON deserialization, and Jinja2 rendering.
- **Werkzeug Security:** Industry-standard cryptographic engine providing password hashing via PBKDF2/Scrypt algorithms with randomized salting.
- **OpenPyXL (v3.1+):** Python spreadsheet library used to programmatically assemble official COE attendance spreadsheets with cell styling, auto-column widths, and metadata banners.
- **Gunicorn (v21.2+):** WSGI server configured with 2-4 concurrent worker threads to service multiple room invigilators simultaneously in university deployments.

4.2 Key Backend Services & Design Patterns

1. **Role-Based Access Control (RBAC):**

   Custom Python decorators `@login_required` and `@admin_required` govern endpoint access. Faculty invigilators have operational access across any exam hall on campus, while configuration endpoints (user accounts, syllabus, session deletion) are locked to the Administrator.

2. **Atomic Attendance Engine (`/api/attendance/mark`):**

   All database operations are wrapped inside immediate database transactions. If any verification fails (such as barcode already assigned, session locked, or invalid candidate), the transaction rolls back cleanly with zero orphaned records.

3. **Indian Standard Time (IST) Window Evaluator:**

   A specialized utility calculates session windows relative to standard Indian Time (UTC+05:30). Sessions strictly transition through `UPCOMING` (pre-exam), `ACTIVE` (window open for scanning), and `CONCLUDED` (closed), with administrative override bypasses available for authorized staff.

4. **Spreadsheet Formula Injection Defense:**

   To eliminate CSV/Excel injection vulnerabilities (where an attacker crafts a USN starting with `=CMD()`), the Excel generator automatically prepends a single quote (`'`) to any string starting with `=` , `+` , `-` , or `@` .

5.0 Relational Database Schema & Data Integrity

The persistence layer is anchored by **SQLite3** configured with Write-Ahead Logging (`PRAGMA journal_mode=WAL`) for superior concurrent read/write performance. The relational schema is structured in Third Normal Form (3NF):

TABLE NAME	PRIMARY ATTRIBUTES	RELATIONAL CONSTRAINTS & INDEXES
`users`	`id, username, email, password_hash, full_name, role, department_id, is_active`	`UNIQUE(username)` , `UNIQUE(email)` , Check `role IN ('ADMIN','TEACHER')`
`departments`	`id, code, name, created_at`	`UNIQUE(code)` (e.g. CSE, ISE, ECE, ME, CV, EEE)
`classrooms`	`id, department_id, room_number, building, capacity`	`UNIQUE(department_id, room_number)` , FK to `departments(id)`
`examinations`	`id, exam_code, exam_name, academic_year, term`	`UNIQUE(exam_code)` (e.g. EXAM-2026-EVEN)
`subjects`	`id, department_id, subject_code, subject_name, qp_code`	`UNIQUE(subject_code)` , FK to `departments(id)`
`exam_sessions`	`id, exam_id, department_id, classroom_id, course_code, room_number, exam_date, start_time, end_time, status`	FK to `classrooms` , `departments` , `users` . Check `status IN ('IN_PROGRESS','LOCKED','SUBMITTED')`
`students_master`	`id, usn, student_name, department_id, semester`	`UNIQUE(usn)` , FK to `departments(id)`
`students`	`id, session_id, student_master_id, usn, student_name, desk_number, attendance_status, qp_barcode`	`UNIQUE(session_id, usn)` , Check `attendance_status IN ('PENDING','PRESENT','ABSENT')`
`booklet_registry`	`id, barcode_value, exam_id, batch_number, status, assigned_session_id, assigned_usn`	`UNIQUE(barcode_value)` , Check `status IN ('AVAILABLE','ASSIGNED','VOID')`
`audit_trail`	`id, session_id, usn, action_type, old_status, new_status, qp_barcode, reason, invigilator_id, timestamp, client_ip`	Immutable transaction log. Stores full historical changes with mandatory justifications.
`evaluation_verifications`	`id, usn, scanned_qp_barcode, session_id, evaluator_name, verification_status, details, timestamp`	Evaluation center cross-checking log to detect paper mismatches or unrecorded scripts.

6.0 State Transitions & Persistence Resilience Engine

6.1 Candidate Attendance State Machine

Every student allocated to an examination hall transitions through strict, mutually exclusive states:

- `PENDING`: Initial default state upon room allocation. Student is unconfirmed.
- `PRESENT`: Transitioned upon valid booklet barcode scan. Booklet becomes `ASSIGNED`.
- `ABSENT`: Explicitly confirmed absent by invigilator. Booklet barcode remains null.
- **REDO Transition:** If a student was marked by mistake, the invigilator executes a redo. The previous barcode is released to `AVAILABLE`, the audit log records the correction reason, and the candidate returns to `PENDING`.

6.2 Persistent Snapshot Engine

Cloud containers (e.g. Render, Docker) often experience ephemeral storage wipes on server restart. To guarantee scheduled exams never vanish:

- **Snapshot Trigger (`save_backup_sessions`):** Executed immediately whenever an exam is created, schedule edited, or students enrolled. Serializes full session metadata and rosters into `scheduled_exams_backup.json`.
- **Auto-Reconciliation (`restore_backup_sessions`):** Executed upon server boot immediately after `init_db()`. Reads the JSON store and re-inserts any missing sessions and student rosters.
- **Permanent Persistence:** Scheduled exams remain safely intact across all logouts and reboots until an Administrator explicitly deletes them.

7.0 Security Architecture & Anti-Fraud Measures

SECURITY THREAT	VULNERABILITY VECTOR	COE EXAMATTEND MITIGATION STRATEGY
Duplicate Booklet Assignment	An invigilator accidentally or intentionally scans the same booklet barcode for two different students.	**Registry Collision Check:** The backend checks `booklet_registry` within a database transaction. If `status == 'ASSIGNED'`, it aborts immediately with `HTTP 409 Conflict`.
Counterfeit Barcodes	A candidate brings in an unauthorized external answer booklet with a fake barcode.	**Closed Registry Validation:** Barcodes must exist in the COE pre-registered inventory table. Unknown barcodes are rejected with `HTTP 400 Bad Request`.
Post-Submission Tampering	An invigilator changes an absent candidate to present after the examination has concluded.	**Session Submission Lock:** Submitting transitions session status to `LOCKED`. All standard update endpoints reject requests with `HTTP 403 Forbidden`.
Privilege Escalation	A teacher attempts to delete an exam or access admin configuration by typing URLs directly.	**Strict RBAC Decorators:** Routes are guarded with `@admin_required`. Teacher attempts trigger an immediate 403 rejection and audit logging.
Paper Evaluation Swapping	A dishonest evaluator replaces a student's answer script during valuation at the evaluation center.	**Cross-Verification Portal (`/evaluator`):** Evaluators scan barcodes; the system verifies against hall records. Unmatched barcodes flag instant fraud warnings.

8.0 End-to-End Operational Workflows

8.1 Master Examination Timetable & Scheduling

1. **Access Master Schedule (`/exams`):** Both teachers and administrators can view the unified institutional timetable. Filter by academic department (CSE, ISE, ECE, ME, CV, EEE) or status.
2. **Admin Scheduling (`/admin/exams/create_session`):** Administrators define the subject code, course name, date, start time, end time, and assign an invigilator.
3. **Handwritten Room Numbers:** Administrators can type any custom room string (e.g. `CS101`, `Room 302`). If the room does not exist, the system creates it on the fly.
4. **Roster Auto-Cloning:** If scheduling consecutive exams in the same hall, candidate rosters automatically carry forward with fresh `PENDING` states.

8.2 Invigilation & Real-Time Barcode Attendance

1. **Login & Hall Selection:** Invigilator logs in with institutional credentials and opens their assigned room from the schedule or department cards.
2. **Barcode Scan:** Invigilator taps **"Scan Booklet Barcode"** on a student card. The phone camera opens, detects the booklet barcode, verifies it, and marks the student `PRESENT`.
3. **Marking Absent:** For absent candidates, the invigilator taps **"Mark Absent"**; the state updates to `ABSENT` without tying up any booklet barcode.
4. **Pre-Submission Review:** At `/sessions/<id>/review`, students are sorted into Present, Absent, and Unmarked. The system displays a stern warning if any candidate remains unverified.
5. **Session Finalization:** The invigilator confirms total counts. The session transitions to `LOCKED`, producing the official COE Excel report.

8.3 Official COE Excel Attendance Report Generation

When an exam is finalized, the system compiles the official Controller of Examinations spreadsheet via `/download_report/<session_id>` adhering strictly to university standards:

- **Header Metadata Block:** Institution name, academic term, department, course code, room number, exam date, and invigilator details.
- **14 Standard Columns:** Sl. No, Desk No, USN, Candidate Name, Attendance Status, Answer Booklet Barcode, Invigilator Initial, Candidate Signature space, and Remarks.
- **Absentee Integrity:** Absent candidates are explicitly marked `ABSENT` with blank barcode cells to prevent phantom booklet submission.
- **Statistical Summary & Signatures:** Footer block with Total Enrolled, Present Count, Absent Count, Completion %, Chief Superintendent, and COE signatures.

9.0 Automated Verification & 30-Scenario Test Suite

The project includes a comprehensive regression test suite (`test_comprehensive.py` and `test_system.py`) covering 30 functional and security criteria. All test cases execute with 100% pass rates:

TEST ID	VERIFICATION SCENARIO & DESCRIPTION	TARGET COMPONENT	RESULT
TEST 01	Valid teacher authentication & session cookie initialization	RBAC / Login	**PASSED**
TEST 02	Invalid credentials rejection with user-friendly feedback	Security / Login	**PASSED**
TEST 03	Department duty routing & access authorization	RBAC / Depts	**PASSED**
TEST 04	Classroom authorization & dynamic session loading	RBAC / Classrooms	**PASSED**
TEST 05	Dynamic student roster loading from relational database	Database / Roster	**PASSED**
TEST 06	Valid registered barcode from booklet registry accepted	Barcode Engine	**PASSED**
TEST 07	Unknown / unregistered barcode rejected with HTTP 400	Anti-Fraud	**PASSED**
TEST 08	Invalid barcode format rejected gracefully	Input Validation	**PASSED**
TEST 09	Barcode already assigned to another student rejected (HTTP 409)	Duplicate Prevention	**PASSED**
TEST 10	Repeated scans do not create duplicate attendance rows	Concurrency	**PASSED**
TEST 11	Failed barcode validation does not mark student present	Data Integrity	**PASSED**
TEST 12	Attendance saved transactionally in SQLite database	SQLite WAL	**PASSED**
TEST 13	Attendance state persists across browser page refreshes	Persistence	**PASSED**
TEST 14	Redo correction updates attendance state correctly	State Machine	**PASSED**
TEST 15	Audit trail retains original and corrected states with reasons	Audit Trail	**PASSED**
TEST 16	Pending / unmarked students identified before submission	Review Module	**PASSED**
TEST 17	Unresolved students cannot silently become confirmed absentees	Validation Rule	**PASSED**
TEST 18	Invigilator can correct errors prior to final submission	UX / Workflow	**PASSED**
TEST 19	Submitted session cannot be modified via ordinary routes	Submission Lock	**PASSED**
TEST 20	Unauthorized unlock attempts rejected with HTTP 403	Security Override	**PASSED**
TEST 21	Authorized COE unlocking requires PIN & mandatory logging	COE Emergency PIN	**PASSED**
TEST 22	Excel report contains all 14 required institutional columns	Excel Generator	**PASSED**
TEST 23	Absent candidates appear correctly with blank barcode cells	Excel Formatting	**PASSED**
TEST 24	Optimistic locking prevents concurrent double-assignment	Concurrency Locks	**PASSED**
TEST 25	Mobile interface maintains responsive viewport & touch targets	Mobile Frontend	**PASSED**
TEST 26	Camera permission denial handled gracefully	WebRTC Scanner	**PASSED**

TEST ID	VERIFICATION SCENARIO & DESCRIPTION	TARGET COMPONENT	RESULT
TEST 27	Camera scanner stops cleanly when dismissed	Camera Lifecycle	**PASSED**
TEST 28	Network errors do not falsely report successful persistence	Error Handling	**PASSED**
TEST 29	CSV bulk student import detects duplicate USNs & row errors	Admin Import	**PASSED**
TEST 30	Faculty cannot bypass authorization to call administrative APIs	Privilege Escalation	**PASSED**

10.0 Deployment Architecture & Operational Setup

10.1 Cloud Deployment (Render / Docker)

The application includes complete configuration files (`Dockerfile`, `render.yaml`, `Procfile`) for zero-downtime containerized cloud deployment:

```
# Production Container Invocation
web: gunicorn wsgi:app --workers 2 --threads 4 --timeout 120 --bind 0.0.0.0:$PORT
```

10.2 Campus Wi-Fi Testing (Local Network Access)

Because the launcher (`run.py`) binds to `0.0.0.0:5000`, invigilators on the university Wi-Fi network can connect their smartphones directly:

1. Connect phone and server computer to the same campus Wi-Fi network.
2. Identify computer IP (e.g. `192.168.1.45`) via `ipconfig`.
3. Navigate to `http://192.168.1.45:5000` in mobile Safari / Chrome.
4. For WebRTC camera compliance over LAN, run `START_PUBLIC_HTTPS_TUNNEL.bat` to obtain a public HTTPS URL.

Pre-Configured Demonstration Accounts:

- **Administrator:** Username: `admin` | Password: `AdminPassword@2026`
- **Faculty (CSE):** Username: `ramesh.cs` | Password: `TeacherPass@2026`
- **Faculty (ISE):** Username: `priya.cs` | Password: `TeacherPass@2026`
- **COE Emergency Unlock PIN:** `COE@2026`

End of Report • Digital Examination Attendance & Verification System (COE ExamAttend v2.5) • Autonomous Institutions